EE2016 Microprocessor Theory and Lab

Tutorial on Memory

1. Interface a 4kbyte memory module to a microprocessor having 20 bit address port and 8 bit data port using high order memory interleaving so that the starting address of the memory map is 80000H.
2. Using 2kbyte memory modules and using high order memory interleaving, realize a memory map of size 4Kword (word = 16 bits) of a microprocessor having 24 bit address port and 16 bit data bit port so that the starting address of the memory map is B00000H.
3. Consider a microprocessor of 16 bit address port and 8 bit data port. If you interface a 1 kbyte memory module, and generate an active high module-select signal using partial decoding by ANDing A15 and A14. how many addresses of the processor's address space will get mapped onto the zeroth address of the memory module and what are they.
4. Consider a low order memory interleaving scheme that uses 4 low order address lines to generate the module select.The 6 high order address lines are connected to the address ports of the memory module. Let T be the individual access time of the memory modules. Find the effective access time for accessing a long sequence of memory addresses for the following cases.
    a. Skip distance of 2 when simplex low order interleaving scheme is used.
    b. Skip distance of 4 when complex low order interleaving scheme is used
    c. Skip distance of 5 when simplex low order interleaving scheme is used
    d. Skip distance of 7 when complex low order interleaving scheme is used
5. Consider a memory interleaving scheme which uses a combination of high and low order memory interleaving as follows. Totally there are “h+n+l” address lines. Memory modules each of size 2n locations are arranged in the form of a two dimensional array with 2h columns and 2l rows. Columns are numbered from 0 to 2h-1. Rows numbered from 2l-1. The higher order h lines are used to generate the 2h column select lines,one for each column. The lower order “*l*” lines are used to generate the 2l row select lines one for each row. Each module gets two select lines; one row select line and one column select line. The middle order n lines go to the address ports of all the memory modules. Derive the general expression for the addresses that get mapped onto module (i,j) located in the i th row and j th column.